<u>Quantum</u> lattice of ions ("matress")

$$\hat{H} = \sum_{j=1}^{N} \left[\frac{\hat{p}_j^2}{2m} + \frac{1}{2} k \left(\hat{x}_j - \hat{x}_{j-1}\right)^2 \right]$$

but $[\hat{x}_j, \hat{p}_n] = i\hbar\, \delta_{jn}$ (position and momentum do not commute only if they describe motion of the same ion #j)

FT
$$\begin{cases} \hat{x}_j = \frac{1}{\sqrt{N}} \sum_q \hat{x}_q \, e^{iqja} \\ \hat{p}_j = \frac{1}{\sqrt{N}} \sum_q \hat{p}_q \, e^{iqja} \end{cases}$$

Inverse FT
$$\begin{cases} \hat{x}_q = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \hat{x}_j \, e^{-iqja} \\ \hat{p}_q = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \hat{p}_j \, e^{-iqja} \end{cases}$$

commutator
$$[\hat{x}_q, \hat{p}_k] = \frac{1}{N} \sum_{j=1}^{N} \sum_{m=1}^{N} \left[\hat{x}_j \, e^{-iqja}, \hat{p}_m \, e^{-ikma} \right] =$$

$$= \frac{1}{N} \sum_{jm} \underbrace{[\hat{x}_j, \hat{p}_m]}_{i\hbar\, \delta_{jm}} e^{-ia(qj+km)} =$$

$$= i\hbar \underbrace{\frac{1}{N} \sum_m e^{-i(q+k)ma}}_{\delta_{q+k,0}} = i\hbar\, \delta_{q+k,0}$$

$$\sum_{j=1}^{N} \hat{P}_j^2 = \frac{1}{N} \sum_{j=1}^{N} \sum_{qq'} \hat{P}_q \hat{P}_{q'} e^{iqja + iq'ja} =$$

$$= \left(\frac{1}{N} \sum_{j=1}^{N} e^{ij(q+q')a} = \delta_{q+q',0} \right) = \sum_q \hat{P}_q \hat{P}_{-q}$$

$$\sum_{j=1}^{N} (x_j - x_{j-1})^2 = \frac{1}{N} \sum_{qq'} \sum_j \hat{x}_q \left(e^{iqja} - e^{iq(j-1)a} \right) \cdot$$

$$\cdot \hat{x}_{q'} \left(e^{iq'ja} - e^{iq'(j-1)a} \right) =$$

$$= \frac{1}{N} \sum_{qq'} \hat{x}_q \hat{x}_{q'} \sum_{j=1}^{N} e^{ijaq} \left(1 - e^{-iqa}\right) e^{ijaq'} \left(1 - e^{-iq'a}\right)$$

$$= \left(\frac{1}{N} \sum_{j=1}^{N} e^{ia(q+q')j} = \cancel{\delta(q+q')} \; \delta_{q+q',0} \right) = \sum_q \hat{x}_q \hat{x}_{-q} \underbrace{\left(1 - e^{-iqa}\right)\left(1 - e^{iqa}\right)}_{1+1-2\cos qa}$$

$$= \sum_q 2(1 - \cos qa) \hat{x}_q \hat{x}_{-q} = \sum_q 4 \sin^2\left(\frac{qa}{2}\right) \hat{x}_q \hat{x}_{-q}$$

$$\Rightarrow \hat{H} = \sum_{q \in BZ} \left\{ \frac{1}{2m} \hat{P}_q \hat{P}_{-q} + k \sin^2\left(\frac{qa}{2}\right) \hat{x}_q \hat{x}_{-q} \right\}$$

$$k = m\omega_0^2$$

$$E_q^2 = 4\omega_0^2 \sin^2 \frac{qa}{2}$$

$$= \sum_q \left\{ \frac{1}{2m} \hat{P}_q \hat{P}_{-q} + \frac{1}{2} m E_q^2 \hat{x}_q \hat{x}_{-q} \right\}$$

$$E_q = 2\omega_0 \left| \sin \frac{qa}{2} \right| = E_{-q}$$

$\Rightarrow \hat{H}$ is (almost) collection of harmonic oscillators at momenta q.

(the difference is $\hat{P}_q \hat{P}_{-q}$ & $\hat{x}_q \hat{x}_{-q}$ instead of $\hat{P}_q^2$ & $\hat{x}_q^2$)

We now introduce

$$\left.\begin{aligned} \hat{x}_q &= \sqrt{\frac{\hbar}{m E_q}}\ \frac{b_q + b^+_{-q}}{\sqrt{2}} \\ \hat{p}_q &= \sqrt{\hbar m E_q}\ \frac{b_q - b^+_{-q}}{i\sqrt{2}} \end{aligned}\right\}$$

check below that all comm. relations are satisfied.

$$\hat{H} = \sum_q \left\{ \frac{\hbar m E_q}{2m} \frac{-1}{2} \left(b_q - b^+_{-q}\right)\left(b_{-q} - b^+_q\right) + \frac{1}{2} m E_q^2 \frac{\hbar}{m E_q} \cdot \frac{1}{2} \cdot \right.$$

$$\left. \cdot \left(b_q + b^+_{-q}\right)\left(b_{-q} + b^+_q\right) \right\} =$$

$$= \sum_q \left\{ \frac{-1}{4} \hbar E_q \left(\underline{b_q b_{-q}} - b_q b^+_q - b^+_{-q} b_{-q} + \underline{\underline{b^+_{-q} b^+_q}} \right) \right.$$

$$\left. + \frac{1}{4} \hbar E_q \left(\underline{b_q b_{-q}} + b_q b^+_q + b^+_{-q} b_{-q} + \underline{\underline{b^+_{-q} b^+_q}} \right) \right\}$$

$$= \sum_q \frac{1}{2} \hbar E_q \left(b_q b^+_q + b^+_{-q} b_{-q} \right) =$$

$$= \sum_q \frac{1}{2} \hbar E_q \underset{q \to -q}{b^+_{-q} b_{-q}} + \sum_q \frac{1}{2} \hbar E_q \left(1 + b^+_q b_q\right) =$$

$$= \sum_q \frac{1}{2} \hbar E_q\, b^+_q b_q + \sum_q \frac{1}{2} \hbar E_q \left(b^+_q b_q + 1\right)$$

$$= \sum_q \hbar E_q \left(b^+_q b_q + \frac{1}{2} \right)$$

$\Rightarrow$ It is just a collection of harmonic oscillators: one for each momentum $q = \frac{\pi n}{Na}$ ($n = 0, \pm 1, \pm 2 \ldots$ inside the first Brillouin zone. <u>PHONONS</u>

Check commutation relations

$$\hat{x}_q + \frac{i\hat{p}_q}{mE_q} = \sqrt{\frac{\hbar}{mE_q}} \underbrace{\left(\frac{b_q + b_{-q}^+}{\sqrt{2}} + \frac{b_q - b_{-q}^+}{\sqrt{2}} \right)}_{\sqrt{2}\, b_q}$$

$$\Rightarrow b_q = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_q + \frac{i}{mE_q} \hat{p}_q \right)$$

$$\cancel{\Rightarrow b_q^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_q - \frac{i}{mE_q} \hat{p}_q \right)}$$

$$\cancel{\Rightarrow b_{-q}^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_{-q} - \frac{i}{mE_q} \hat{p}_{-q} \right)}$$

This can be checked as

$$\hat{x}_{+q} - \frac{i}{mE_q} \hat{p}_{+q} = \sqrt{\frac{\hbar}{mE_q}} \underbrace{\left(\frac{b_{+q} + b_{-q}^+}{\sqrt{2}} - \frac{b_{+q} - b_{-q}^+}{\sqrt{2}} \right)}_{\sqrt{2}\, b_{-q}^+}$$

$$\Rightarrow b_{-q}^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_q - \frac{i}{mE_q} \hat{p}_q \right)$$

thus $b_q^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_{-q} - \frac{i}{mE_q} \hat{p}_{-q} \right)$

! Note that $\left[\hat{b}_q, \hat{b}_k^+ \right] = \sqrt{\frac{mE_q}{2\hbar}} \sqrt{\frac{mE_k}{2\hbar}} \left[\hat{x}_q + \frac{i}{mE_q} \hat{p}_q, \right.$

$$\left. \hat{x}_{-k} - \frac{i}{mE_k} \hat{p}_{-k} \right] = \frac{m}{2\hbar} \sqrt{E_q E_k} \left(\frac{-i}{mE_k} \underbrace{\left[\hat{x}_q, \hat{p}_{-k} \right]}_{i\hbar\, \delta_{q+k,0}} + \frac{i}{mE_q} \underbrace{\left[\hat{p}_q, \hat{x}_{-k} \right]}_{-i\hbar\, \delta_{q-k,0}} \right)$$

$$= \frac{m}{2\hbar} E_k \frac{2\hbar}{mE_k} \delta_{q-k,0} = \delta_{q-k,0}$$

This is how it must be because

$$\hat{x}_q = \frac{1}{\sqrt{N}} \sum_j \hat{x}_j \, e^{-iqja}$$

but $$\left(\hat{x}_q\right)^+ = \frac{1}{\sqrt{N}} \sum_j \underset{\hat{x}_j}{\underset{\parallel}{\hat{x}_j^+}} \, e^{iqja} = \frac{1}{\sqrt{N}} \sum_j \hat{x}_j \, e^{-i(-q)ja} = \hat{x}_{-q}$$

The same for momentum: $\hat{p}_q^+ = \hat{p}_{-q}$

Thus, while in position space $\hat{x}$ and $\hat{p}$ are Hermitian (self-adjoint) operators, in momentum space they are not!